

A Packing of 93 Lines in $\mathbb{R}^{11}$

Limin Ge*

October 8, 2026

Abstract

We give 93 1-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 0.832245. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.831003, obtained by deleting 1 of the 94 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 93 subspaces of dimension 1 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.832245 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 1$. The cell $N = 93$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.831003, obtained by deleting 1 of the 94 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(11 - 1)/11 \cdot 93/92 = 0.918972$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n1N93.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 78 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:  
832245912953385410891838810280102366402058164723  
denominator:  
999999999999080161821573063943174484199850386532
```

B Coordinates

93 blocks of 11 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m11n1N93.txt`.

966728240	50150137865	-83896244551	328834127141
363016091121	212667334289	158950617445	429872331635
357188582307	-168312368790	-265887561716	281369958787
113926749743	-301832754300	-458938748372	371928612242
-77245368960	172180728752	144320613639	569671431393
431996986134	-404255218365	11173399997	329830241848
-405560767301	-382700211813	-333608697914	-103802506303
-328837944672	332268265512	-67508869076	322049655650
-55378420463	-624794480376	227871163665	-258427362347
-97226836729	-141691727998	-649803293711	-379308558751
-499853906273	91008770704	-136669926203	-531414999648
-233685140727	-426120174122	-247475863994	-123465167791
-36707447980	357619871103	-290474164219	-113581670209
-647726654313	392372486432	118751548673	-17711206511
-80897227012	-64985836747	-42727241413	137530073237
283137154315	-48764485350	-446459556076	-325745020915
193292215	-28666055308	641894386	-258669074190
478204190223	4592956345	96054575884	147100997183
-281171373952	-395077128848	180953829038	676436125489
-313528437613	-585192748809	71174851859	125412954554
-75079588471	-52725458850	292175225546	-47675924760
-161583841490	-367322503752	329883292541	521704753067
-13413842794	-323883478815	-590923584424	-182896126000
-488812735573	-445790900639	-390424470892	-18325600926
302102713546	151726777989	236503531462	471435015113
234268807393	-98129235145	71170210952	-324777225992
333687154637	91607627367	121577141504	-39519047408
-15392839369	-13966152661	559004262913	-146311307888
-376228146518	-137442007866	508022010839	-346544981184
-46884513548	-404343745271	-120331834974	437347532687
277307084300	131480208928	-301243722429	151680993879
147729020916	-161584746015	207132879670	94678106842
-510569622651	30651856541	122441285222	294945790733
-55253199325	-465148987526	-208071198292	382875969175
38130924526	77088939091	355645924170	-307800088098
-37167995123	-458327831679	276912611432	-172743636094
128923477563	355622553701	156768136166	524695616665
-273981296378	103355902151	-606100474324	-120721524434
-132256762069	1209398613	-413603454726	-142526220734
99497999415	-471855837738	16291778891	23023677232
-531652962235	-198289629420	-30444253145	-479118716022
9391533479	-96288462420	-134234014701	-206012516444
318963216276	-163576730115	-60756594400	-243131289265
700394412723	310557708377	-58554562076	-53119530063
-193126556556	481063124502	186785161454	-467635405852
-112388734582	-506987604802	592913982415	-390361455296
-96587779764	227869158101	-155664433469	268349813853
485662083734	-275281208405	20616585016	-113220699704
-135569333179	-14348912961	403724653131	-77973900618
-312150320987	449864847041	46169097244	39759527480
-274134379029	97261441497	276509370687	1736539397
640544308622	-421559602534	426033900874	-71156333542
101966665376	176551704205	-202318373038	-594173874748
-122391086201	-73632583233	278594538655	-418943009839
295752656537	251069823669	160689408528	163901614750
-340001497854	138025660230	-169790977171	70527951343
-38752763089	503778565082	169344905171	-682914720806
563800128927	-151605765359	-165805883501	-153206

-138240532622	-768879241484	220472018729	-359041908659
-20376821965	433919345781	191680456438	75188255740
-496868290106	-226373762118	-14643532521	514211668413
-177705569068	58394416873	-525111786178	84286488304
50381368645	-523530598258	45743944285	-51630905896
68843238874	-107686911597	-630645563579	350344203768
53288254814	-342873319502	-46852196168	141459301766
340256908307	328150349380	-195655580364	-85324836573
-470525510516	-273951727003	-37863812715	-327350383339
-227952612169	-58202432689	2106553276	-187362030995
114980763573	-303820971705	443816623118	224004728367
-298130937235	-534773852955	-111783458997	15556274468
-236320782099	167713996468	24273699879	262864351035
2894376471	28447419019	23568963883	-21453670363
-157756078681	-293847173986	182097977846	-611951579822
651924725497	145085044549	-112207977484	269504225690
-13326647590	123001674885	131153119961	-75871736063
61992787361	464019080086	12643453080	473436096224
-38445351686	282976554489	550648996569	246462231577
101020856937	96547209458	364373124861	39556649838
-63468206627	-385895714907	682941758906	-220896276206
121707521195	257270245223	-144419218255	516601350098
577567751075	-545952233105	600883491960	369362809492
-119211533364	-222646020910	144467021475	-173326721982
689706898935	-100126830149	245059453993	-12911393702
-330667606890	-261966709870	-160378289367	228926301405
169198078445	-1060262315	359057408377	-597845323616
64592124450	202567656755	38077928573	-82087370041
317988253634	-132671143460	-126776434033	71861172297
-73517097575	315566267565	253628934472	-307240562368
-147569931899	-65490292472	135693673769	257340353905
-601000939537	-390186630563	431402653879	437588257075
-324714715573	-727069728438	-342052484931	81065714664
-143450800996	27162963844	-93831521182	-104831281652
-473632041139	283208421489	-341557696237	-246403788656
228564573929	83935391653	-90988980398	-333687696754
214086861031	250276954001	393291020151	-436353735371
185792147051	-297355226982	568979192530	430345215355
-166014858230	-415467532915	63706099343	-354943615679
-98515318563	193418194856	246525105076	-203498308394
276173044774	-261982398296	394059941526	98340160687
465600628479	398754873579	360352314961	-22567809729
61346157005	-387414339614	-194460204538	-208316420921
166611220539	408613511878	7412763870	-70137642251
438287588636	299012111200	-299156640432	385264312085
391868284845	24508781231	-68991250150	-244121043295
9086658622	64910896051	-558106154042	-78519264431
-316310638372	-594974179128	-122105207845	-396052015073
374967076717	173372277483	-241637094098	-475316393084
-281646103057	69504372077	369398259189	-586218426743
585538704638	-159680274863	407875599292	3949885458
-100925947930	-154941928396	190471700414	102646029734
-109123511856	601380022547	376615016972	-153548872792
37509143769	-81024378554	-20592978579	-401782353434
-263699988758	-148115694182	199009740608	235819754124
-136252346491	227657953739	330915640553	-267636672951
-30081951090	337832043934	81585207751	-100225463949
440677119803	-74110433592	-260187056818	360580815813
-202859815777	302243626847	-423854550258	-35537963660
485079703324	-102721004846	-236144307975	-195611845047
272002655448	-370745253959	612502183372	-507309973605
257770637299	479168784602	-53501110714	-289281461657
-13304275679	470833123384	-46802188804	-408797696265
-145519751913	-140678292899	-47337742655	-333890828703
131111236366	-394773055590	326083340013	450264698203
301561787263	255899111862	303218781313	-533017446046
182903636166	-97447595378	-103745678020	254808759954
-315520665288	241318062770	444857699490	328033664644
272900725399	22530943604	195980196880	184313474293
7546460826	-142883288894	-72646464763	-65324940376
-756754413287	-59531369531	-371574617560	182508177540
154452586219	-281213193635	-559724927100	167259139210
-76947046890	-63602285582	157958405317	-182072747433
-20256270482	380227629389	330413428922	-310585389973
364514083022	434041649156	210084227647	-190312008904
-55724789580	-419100374048	150451660011	15423822167
894742138101	-51403376517	-308865744497	-650893639258
183869964274	-291808563698	293312259215	132102825277
-76879341689	345496960401	167349414877	91213006518
-23076981473	-420881514562	98756007594	182383963841
-127055909433	-192543975173	767419525135	-353717989523
-13866006942	276375764849	-283815254297	-353903495672
-23301472205	18487471065	-31954777694	336661248877
-357841174560	-366543262749	51880475641	232875916070
-319562287362	-376179767289	22274186199	250818553815
-507747896761	21555900719	-183938842740	94778041083
114293239370	-244153482426	-391949894272	-61758503497
103783049323	8984016941	-122712950982	571207221179
-38694408683	485638399579	109394368998	-86211364375
52881524396	-86115675859	145090729594	207714237591
579377135639	554003095378	-100387781995	-498944746947
-242443444935	-31347399737	12786392489	112086754637
172493302514	193196711124	301234251956	-131552128848
-244302075588	74724587911	5581302911	493251059319
-98154758766	-146273302343	-167550168804	296836312431
-114789511	354623276018	71577348261	-11137738968
150857019173	434462166329	-630324031246	26186221135
-79966692957	-559949206843	190888592323	-420134761388
-138835348286	230033733858	631683396348	127647729625
-169769263094	495899261654	132547250186	-323912126745
-103254200548	-27193976873	-269433329356	439253949735
267859038637	-84163313951	592500477749	-154893471180

466786200579	-438882397656	668523151246	259965077194
-103653608566	302828456270	34076394999	-15568637805
48040832293	-276808479584	-230027539315	187654447090
-390974164725	156714217849	369003018204	-80694583431
-317568715279	429869896659	-17406781675	22097674491
-185032734698	149195977718	80178794929	127536428591
-136494569381	-301907275790	-199235430841	-20501950978
256530873536	456200544956	-163041473370	-429929183961
-498943016324	-147615358572	137733686293	183092112036
-388419490218	459760278545	-28204289845	401652714448
-59542851131	39217676120	-460242858132	-465768679907
163260194524	-50143787349	-316600932137	-267218623701
599031113054	-51242963820	-637251520649	113227612705
9194906865	77116222038	-466097143768	316560151784
202123082893	75228280657	-664841052670	-78961641529
227011878606	-786285075010	10077445166	160986524808
143366188861	-247033765297	-185587490597	-431699272941
407966373311	-11786563276	66686407399	-204707529403
156374734944	-2675915110	378681934202	-44677779090
-579478527202	301529483655	-193432444031	-394034972674
79221164699	-45819218483	260403628485	139159869903
-239230329568	-212619906537	180276751654	-379004266546
-41375091429	-173732545684	-137834751083	189676073261
-124204300603	-465410491125	40301185354	-27813564904
202261410502	-239487757575	9919116324	-532622058293
-419203709712	368355311568	-126032726155	-44389357872
-394301882454	258335089464	85056737549	176500346212
-316489848959	214300227195	107798119539	-533165836184
134957049048	5639976000	-373108292237	-85770515468
270582171942	321326928613	-675324167789	113839826084
239087706302	-544902762050	-436248491146	-348386468257
-614435298533	120555243413	-93259936418	23051585714
299016171041	570779196756	42245564178	557423852929
203710221305	-365312176694	400395641530	372272673178
274295782726	30309033209	93692215990	263454799536
-400368271501	-319782226422	-278655675797	171147043779
-54375745725	38786910316	162804692732	42918905042
65682687826	-341169235289	-49160156201	494257012634
352652050459	-284028385313	-218087514196	252268392500
-597309299263	109693642826	436495180222	46101453830
189074074008	-460830382401	-168280627653	177245666206
-147654790263	-8655273661	176897629301	-41607695216
-83050872875	-527317916577	-105074516670	355778049997
-213466380860	-107465390271	202634442546	-333350698290
66209006928	268220768512	295844232914	667034316537
415430471283	-144937202183	-675353534679	136598659594
453142612958	274689317792	-130662591714	-453765331214
-160874764588	-115056358044	310827014271	43473284867
-25628497814	318449178750	46529808663	-127335774086
-488786598210	-131771208119	570195009411	198558418716
-220653175932	305700234415	8560281853	-524828146969
-368946037967	-84316564126	-221084999889	382392497449
8511203946	556246525954	-223560479802	425873502521
563679147130	182019810977	-58169693869	248372932602
-195013796466	-563399231800	-105137698153	-379388524321
-260886064093	47321101085	461980860217	-194211261153
626509626	-6972836489	-481903906064	-171734306566
-124730895139	-58628756482	412804738796	-330998349612
-252176131282	-116139603555	151164319677	92387516878
270608809490	-598324837032	-400632610994	16766984296
57053496216	322462622257	129722293374	75836728336
-363693780946	-349713834927	89831246010	768033261763
224172489230	-185001871859	493301503805	343581201063
35953806853	-108333910062	179827483977	-113647812734
-447382648858	178938818517	-117510300163	-247319783073
412127411022	54342768775	409416753727	-60586615906
-398492064571	256946553387	190008253807	14636060868
-55549409631	74098155985	-357937701133	41002439677
-154696358206	-109964691911	-154413059173	190678420677
477438555541	445601769217	134480565282	275398569903
-170778551654	578456509520	337474033072	-226795590435
365580752932	271451777256	548160767182	220882190413
234188600771	116947658970	267044506013	263246420761
-21149447343	133644316848	232030923605	165205688042
287587372580	492977651977	-112627526600	-7263196211
41896237611	402193546270	-186748612263	-507450699082
-372039981509	19091157281	476028979520	142255865114
-407030726181	777862498	343606748431	-152959749077
211227794933	260833374458	162241568352	579693889333
441117293683	179477106534	579188738879	179135911589
-244149176958	196041890117	-391082783317	-301244127271
-351708818477	-15557384644	393563489824	378591534633
51345938336	-431795691922	431381741644	-1328953852
289402278796	-253816496850	210495498327	